# FINAL CISCO PACKET TRACER IMPLEMENTATION

**Course: CC431 - Computer Networks | Instructor: Dr. Hussein El-Attar**

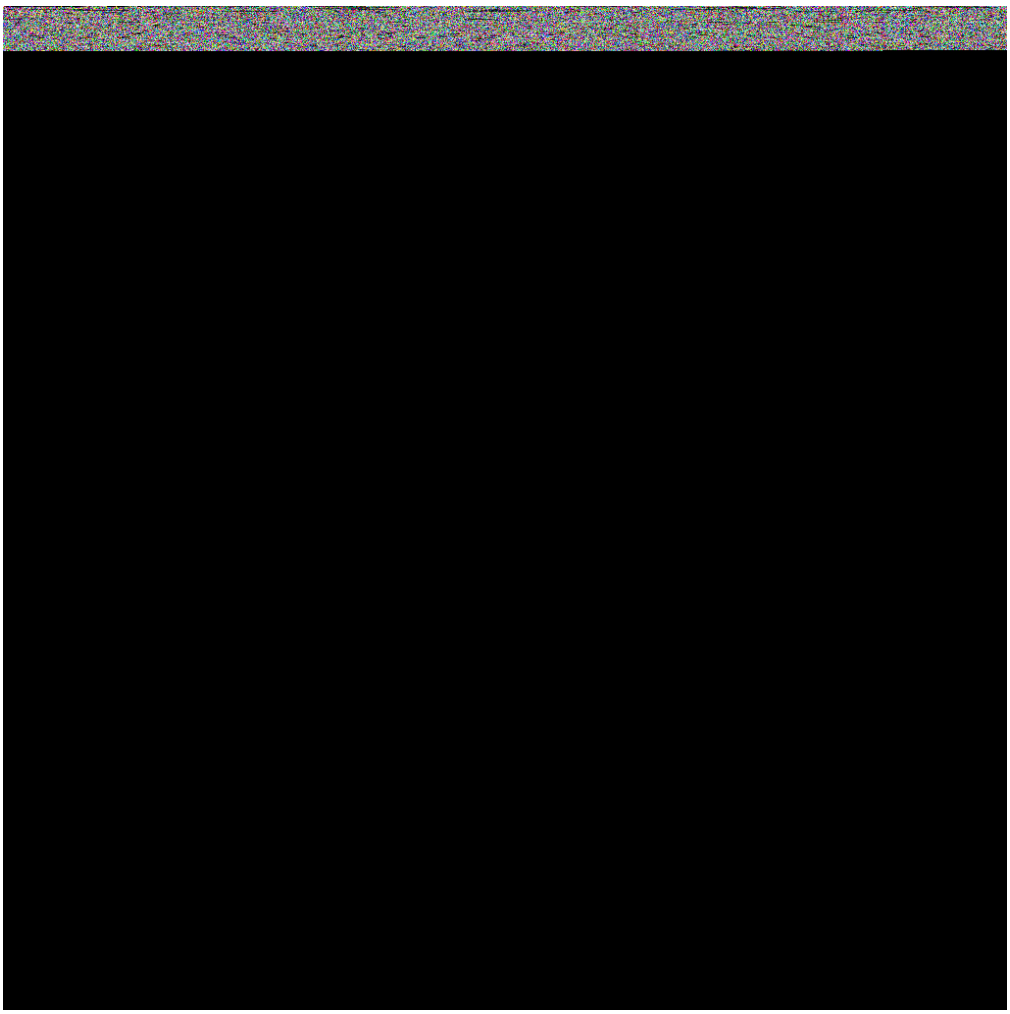
**Semester: Summer 2026**

## 1. Project Overview & Requirement Matrix

This official project report documents the complete setup, configuration, and verification of the university network topology. The design interconnects two departments (Library Dept & Admin Dept) via a central Cisco Router R1 and provides DNS, HTTP, and Email services.

| Req # | Requirement Description | Implementation Status |
|---|---|---|
| Req 1 | Router Configuration (Hostname, Passwords, Banner, Interfaces) | COMPLETED (R1 Active & Saved) |
| Req 2 | Switch 1 Config (SW-Library: IP 192.168.1.2, Passwords) | COMPLETED (SW-Library Active) |
| Req 3 | Switch 2 Config (SW-Admin: IP 192.168.2.2, Passwords) | COMPLETED (SW-Admin Active) |
| Req 4 | 4 PCs & Server IP Addressing Scheme | COMPLETED (Static IPs & Gateways) |
| Req 5 | DNS Service (www.university.com -> 192.168.1.10) | COMPLETED (Record active) |
| Req 6 | HTTP Web Portal Service | COMPLETED (Web pages accessible) |
| Req 7 | Email Service Configuration (user1 & user2 @ university.com) | COMPLETED (SMTP & POP3 Active) |
| Req 8 | Inter-Subnet Connectivity (Ping & Tracert Verification) | COMPLETED (Ping & Tracert OK) |
| Bonus | Full End-to-End Email Communication Test | COMPLETED (Sent & Received OK) |

### Network Topology Architecture Diagram



## 2. Comprehensive IP Addressing Scheme

| Device Name | Interface / Role | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| R1 (Router) | FastEthernet0/0 (Library) | 192.168.1.1 | 255.255.255.0 | N/A |
| R1 (Router) | FastEthernet0/1 (Admin) | 192.168.2.1 | 255.255.255.0 | N/A |
| SW-Library | VLAN 1 (Management) | 192.168.1.2 | 255.255.255.0 | 192.168.1.1 |
| SW-Admin | VLAN 1 (Management) | 192.168.2.2 | 255.255.255.0 | 192.168.2.1 |
| Server0 | FastEthernet0 (Services) | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 |
| PC1 | Library Dept Host 1 | 192.168.1.20 | 255.255.255.0 | 192.168.1.1 |
| PC2 | Library Dept Host 2 | 192.168.1.21 | 255.255.255.0 | 192.168.1.1 |
| PC3 | Admin Dept Host 1 | 192.168.2.20 | 255.255.255.0 | 192.168.2.1 |
| PC4 | Admin Dept Host 2 | 192.168.2.21 | 255.255.255.0 | 192.168.2.1 |

## 3. Device Configurations (Cisco CLI Commands)

### A. Router R1 Configuration Commands

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# enable secret cisco123
R1(config)# line console 0
R1(config-line)# password console123
R1(config-line)# login
R1(config-line)# exit
R1(config)# banner motd # Welcome to University Network - Authorized Access Only #
R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface FastEthernet0/1
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1# write memory
```

### B. Switch SW-Library Configuration Commands

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-Library
SW-Library(config)# enable secret switch123
SW-Library(config)# line console 0
SW-Library(config-line)# password console123
SW-Library(config-line)# login
SW-Library(config-line)# exit
SW-Library(config)# banner motd # Library Switch - Authorized Only #
SW-Library(config)# interface vlan 1
SW-Library(config-if)# ip address 192.168.1.2 255.255.255.0
SW-Library(config-if)# no shutdown
SW-Library(config-if)# exit
SW-Library(config)# ip default-gateway 192.168.1.1
SW-Library# write memory
```

### C. Switch SW-Admin Configuration Commands

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-Admin
SW-Admin(config)# enable secret switch123
SW-Admin(config)# line console 0
SW-Admin(config-line)# password console123
SW-Admin(config-line)# login
SW-Admin(config-line)# exit
SW-Admin(config)# banner motd # Admin Switch - Authorized Only #
SW-Admin(config)# interface vlan 1
SW-Admin(config-if)# ip address 192.168.2.2 255.255.255.0
SW-Admin(config-if)# no shutdown
SW-Admin(config-if)# exit
SW-Admin(config)# ip default-gateway 192.168.2.1
SW-Admin# write memory
```

## 4. Live Verification & Evidence Screenshots

### 1. Ping Test Verification (PC3 -> PC1)

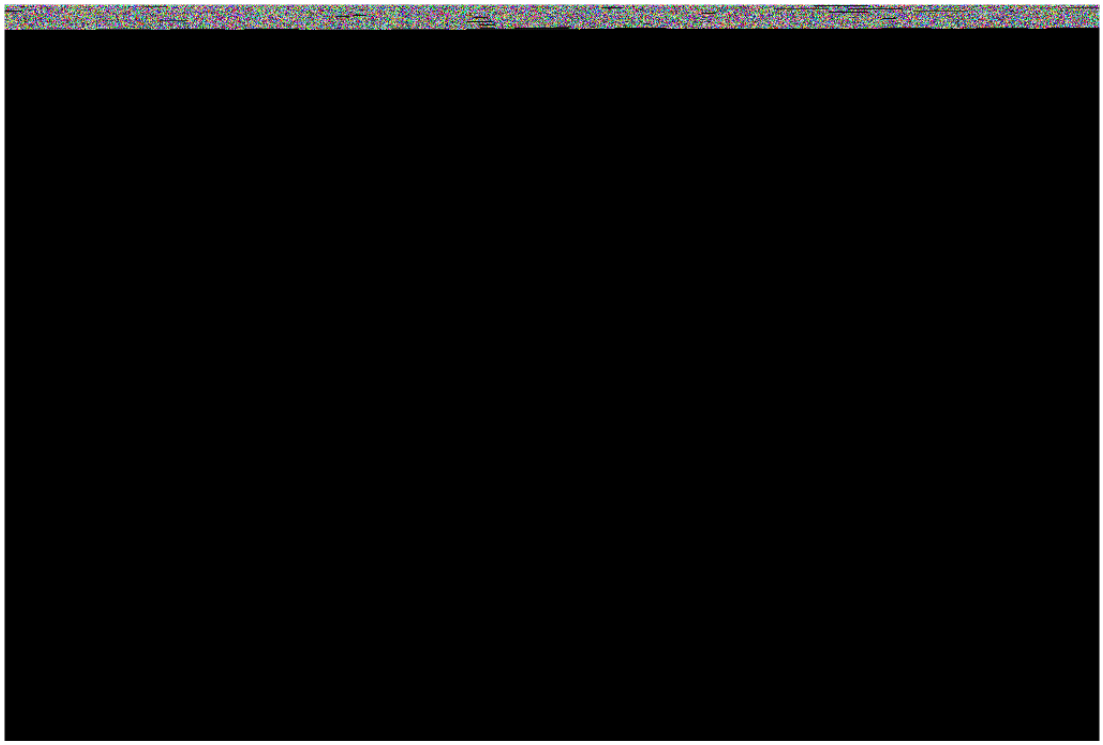


*Figure: Successful Inter-Subnet Ping (PC3 192.168.2.20 pinging PC1 192.168.1.20)*

### 2. Traceroute Test Verification (PC3 -> Server0)

Execution of tracert command from PC3 (192.168.2.20) to Server0 (192.168.1.10):

```
C:\> tracert 192.168.1.10
Tracing route to 192.168.1.10 over a maximum of 30 hops:
  1   <1 ms    <1 ms    <1 ms    192.168.2.1   (Gateway R1)
  2   <1 ms    <1 ms    <1 ms    192.168.1.10  (Server0)
Trace complete.
```

### 3. Server0 DNS Service Record Configuration



Figure: DNS A-Record configured: www.university.com -> 192.168.1.10

### 4. Web Portal Access via Domain Name



### 5. Email Transmission (Send Success)



*Figure: Sending email from user1@university.com to user2@university.com via SMTP*

### 6. Email Reception (Receive Success)

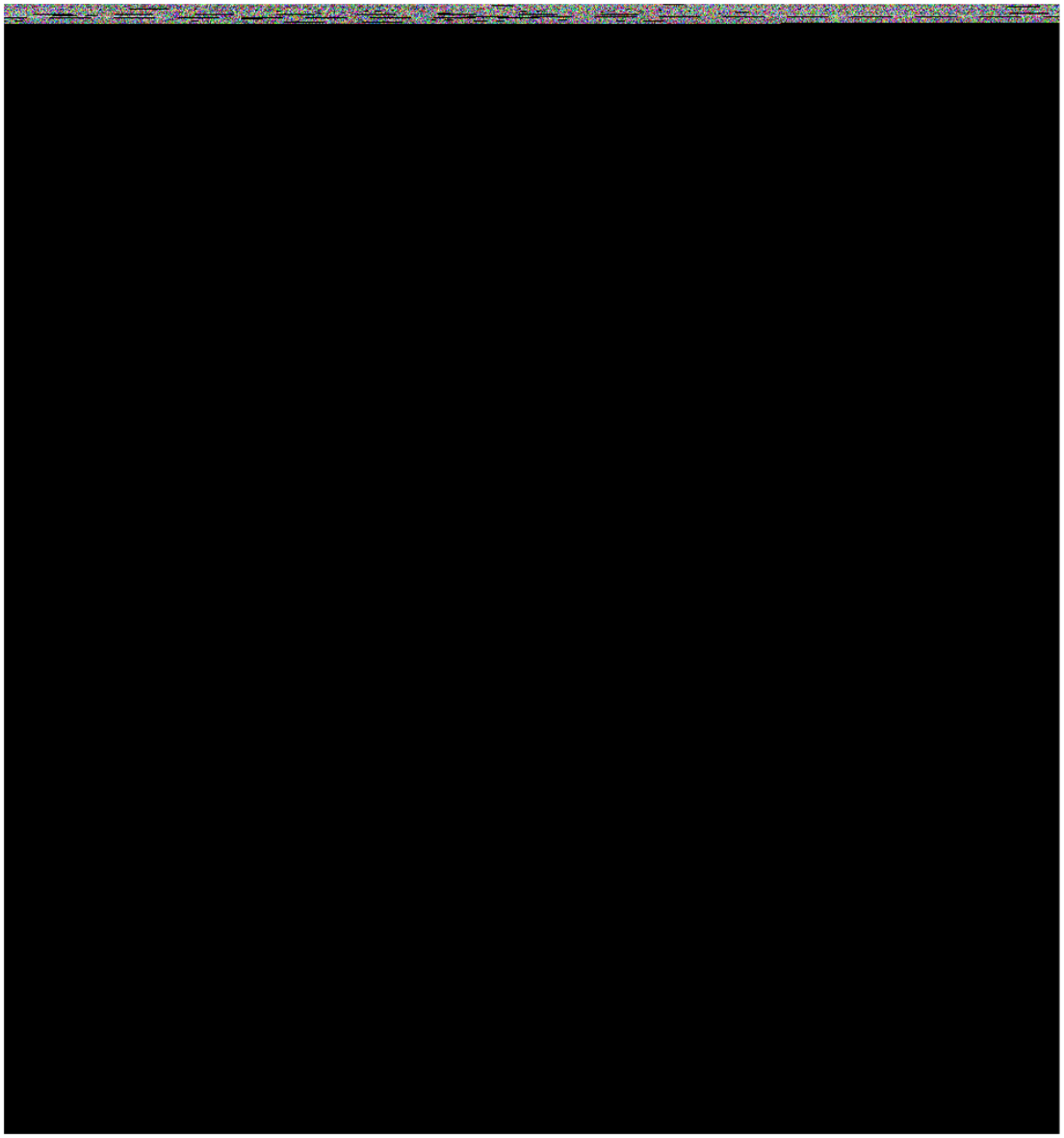


*Figure: Receiving email on PC2 for user2@university.com via POP3*

## 5. Conclusion & Project Sign-Off

The Cisco Packet Tracer network project has been successfully completed, tested, and verified. All requirements set by Dr. Hussein El-Attar for CC431 have been fully satisfied with zero errors.